ORLANDO HEALTH
1414 Kuhl Ave.
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DEPARTMENT GUIDELINE

Guideline Title: **Pediatric Hypotension**	Guideline Number: **GUID-3203-PED009**

Issue date: 09/09/2014	Replaces Dept. Policy:
Revision dates: 4/30/24	Developed by: Air Care Team
Approved by: Dr. Christopher Hunter, MD Air Care Team Medical Director	Approved by:
Signature:	Signature:
Department Numbers: 3203	

I. **PURPOSE:** To recognize the patient with hypotension. To provide and maintain adequate oxygenation, ventilation, and tissue perfusion. To provide adequate blood pressure support and expeditious transport to the nearest appropriate medical facility.

II. **DEFINITIONS:** Hypotension in children is determined by age & systolic blood pressure (BP) mmHg.

A. Term neonates (0 to 28 days): The systolic BP is < 60 mmHg
B. Infants (1 to 12 months): Systolic BP is < 70 mmHg.
C. Children 1 to 10 years (5th BP percentile): Systolic BP is < 70 mmHg + (age in years x 2).
D. Children > 10 years: Systolic BP is < 90 mmHg.

III. **DEPARTMENT GUIDELINE:**

A. Assess airway, breathing, and circulation as per *GUID3203-PED001 – Pediatric General Management*.
B. Apply cardiac monitor, monitor pulse, blood pressure, and oximetry. Assess vital signs.
C. Assess for clinical signs: tachypnea, tachycardia, altered mental status, and either bounding pulses in hyperemic extremities (warm shock) or peripheral vasoconstriction with weak peripheral pulses, mottling, and delayed capillary refill in cool extremities (cold shock).
D. Use a color-coded length-based tape to determine the child's weight (if not known)
E. Obtain IV/IO access for the patient's size.
F. Treat underlying cause of hypotension: hypovolemia, decreased vascular resistance, heart failure
G. Fluid resuscitation
   i. Administer 0.9% NS or Lactated Ringers 20 ml/kg IV, may be repeated per *GUID3203-PR015 – Intravenous Fluid Therapy*.
   ii. Most hypovolemic children will require at least 40-60 ml/kg in the first hour and those with sepsis may require up to 80 ml/kg in the first hour.
   iii. To help deliver fluid rapidly: Place an in-line 3-way stopcock in the IV tubing system. Deliver fluid by using a 30- to 60-mL syringe to push fluids through the stopcock or use a pressure bag (beware of risk of air embolism) or a rapid infusion device.
H. If actively hemorrhaging, control bleeding, and consider infusing blood products per *GUID3203-PR002 - Blood Product Administration* or *GUID3203-PR025 – Emergency Blood Product Administration* at 10 ml/kg.
I. Utilize fluid warmer when possible per *GUID3203-PR026 Warming of Blood Products & Intravenous Fluids*.

J. If hypotension persists after adequate fluid resuscitation, initiate one of the following:
    i. **Epinephrine infusion at 0.05 -2 mcg/kg/min IV/IO titrate q 5 min to appropriate SBP**
    ii. **Norepinephrine infusion at 0.03 - 0.5 mcg/kg/min IV/IO titrate q 5 min to appropriate SBP**
    iii. **Dopamine infusion at 5 - 20 mcg/kg/min IV/IO titrate q 5 min to appropriate SBP**
K. If hypotension attributed to heart failure, refer to *GUID3203-PED006 – Cardiogenic Shock*.
L. For patients with hypotension due to a dysrhythmia, follow appropriate guideline.
M. A child in septic shock with fluid-refractory and epinephrine- or norepinephrine-resistant shock may have adrenal insufficiency. If you suspect adrenal insufficiency, or a patient is at known risk for adrenal insufficiency (i.e., history of steroid use), consider **Hydrocortisone 1-2 mg/kg IV** early from a sending facility. If possible, obtain a baseline cortisol concentration before administration, and obtain expert consultation for any additional evaluation and management.
N. Maintain the optimal body temperature.

IV. **DOCUMENTATION:**
    A. EMS Charts

V. **REFERENCES:**
    A. Pediatric Advanced Life Support Manual, 2020. American Heart Association.